

Research Paper

RESEARCH PAPER

Title:

Healthcare Capacity Analytics and Forecasting for Humanitarian Care Systems

1. Introduction

The Unaccompanied Alien Children (UAC) Program operates as a critical humanitarian care pipeline managed by the U.S. Department of Health and Human Services (HHS). The program oversees intake, sheltering, healthcare support, and reunification services for unaccompanied children transferred from U.S. Customs and Border Protection (CBP).

Effective management of this system requires continuous monitoring of operational care load, discharge effectiveness, intake pressure, and healthcare system sustainability.

2. Problem Statement

Although operational data is collected daily, there is limited centralized analytical infrastructure available for evaluating:

- Total system care load
- Inflow and outflow balance
- Capacity strain
- Forecasted healthcare demand
- Operational bottlenecks

This limitation increases the risk of overcrowding, delayed services, and inefficient resource allocation.

3. Project Objectives

Primary Objectives:

- Quantify total healthcare system load
- Analyze operational intake pressure
- Identify prolonged strain periods
- Forecast future care capacity trends

Secondary Objectives:

- Improve situational awareness
- Support policy planning
- Enable data-driven humanitarian operations

4. Dataset Description

The dataset includes:

- Daily apprehensions into CBP custody
- CBP custody load
- Transfers into HHS care
- Active HHS care load
- Discharges and reunification outcomes

Timeframe:
2023–2025 operational records

5. Methodology

5.1 Data Cleaning

- Removal of formatting inconsistencies
- Date standardization
- Missing value handling
- Numeric conversion

5.2 Feature Engineering

Derived metrics include:

- Total System Load
- Net Intake Pressure
- Growth Rate
- Rolling Average
- Forecasted Load Trends

5.3 Exploratory Data Analysis

Time-series visualizations were used to analyze:

- Healthcare capacity trends
- CBP vs HHS operational distribution
- Intake pressure fluctuations
- Rolling averages and backlog indicators

5.4 Forecasting

Facebook Prophet was used to forecast future healthcare system load trends over a 30-day horizon.

6. Dashboard Development

The dashboard was developed using:

- Streamlit for web application development
- Plotly for interactive visualizations
- Prophet for forecasting
- Pandas and NumPy for data analysis

Key dashboard modules:

- KPI Monitoring
- Operational Load Analytics
- Pressure Monitoring
- Forecasting Dashboard
- Policy Recommendations

7. Key Insights

- HHS care load consistently exceeded CBP custody levels.
- Periods of elevated Net Intake Pressure correlated with increased operational stress.
- Rolling averages revealed temporary backlog accumulation periods.
- Forecasting models suggested moderate stabilization in future system load.

8. Recommendations

- Improve discharge efficiency mechanisms
- Enhance predictive monitoring systems
- Increase temporary care infrastructure during surge periods
- Strengthen operational coordination between CBP and HHS

9. Conclusion

This project demonstrates the value of healthcare analytics and forecasting systems in improving humanitarian operational intelligence. By transforming raw operational data into actionable insights, the platform supports informed decision-making, healthcare sustainability, and strategic humanitarian planning.